Practical No. 5B

Aim: Programs on Dialogs. (DatePicker Dialog, TimePicker Dialog, AlertDialog).

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout android:layout_height="match_parent"
   android:layout_width="match_parent"
   xmlns:android="http://schemas.android.com/apk/res/android">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/tv1"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="50dp"
      android:text="DatePicker Dialog"
      android:textSize="20dp" />
   <Button
      android:id="@+id/btnPickDate"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_centerHorizontal="true"
      android:layout_below="@id/tv1"
      android:layout_marginTop="10dp"
      android:text="Pick Date" />
   <TextView
      android:id="@+id/tvDate"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_below="@id/btnPickDate"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="10dp"
      android:textSize="20dp"/>
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/tv2"
      android:layout_centerHorizontal="true"
      android:layout_below="@id/tvDate"
      android:layout_marginTop="50dp"
      android:text="TimePicker Dialog"
      android:textSize="20dp" />
   <Button
      android:id="@+id/btnPickTime"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_centerHorizontal="true"
      android:layout_below="@id/tv2"
      android:layout_marginTop="10dp"
      android:text="Pick Time" />
```

```
    <TextView
        android:id="@+id/tvTime"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/btnPickTime"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="10dp"
        android:textSize="20dp"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/tv3"
        android:layout_centerHorizontal="true"
        android:layout_below="@id/tvTime"
        android:layout_marginTop="50dp"
        android:text="Alert Dialog"
        android:textSize="20dp" />
    <Button
        android:id="@+id/btnClickMe"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_below="@id/tv3"
        android:layout_marginTop="10dp"
        android:text="Show Alert Dialog Window" />
    <TextView
        android:id="@+id/tvAlertDialog"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/btnClickMe"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="10dp"
        android:textSize="20dp"/>

</RelativeLayout>
```

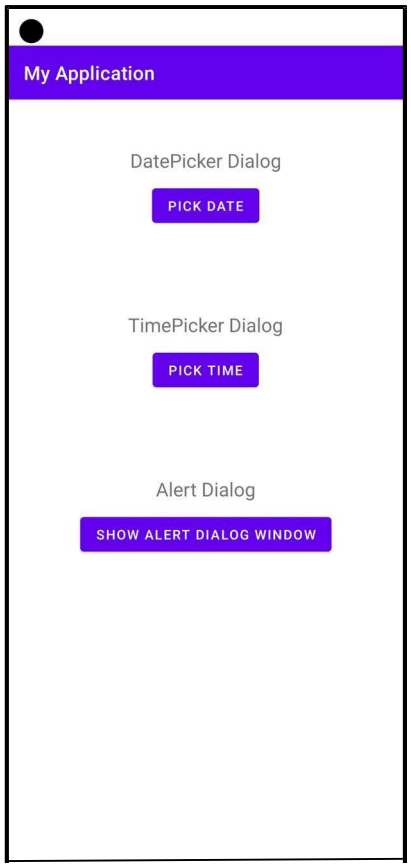
MainActivity.kt

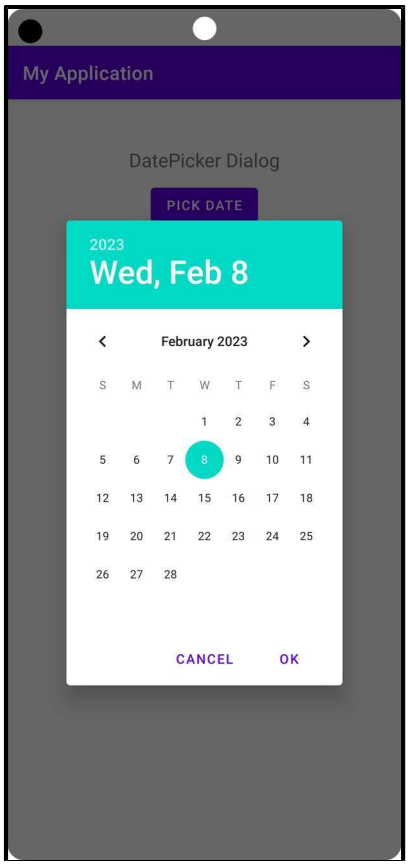
```
package com.example.myapplication

import android.annotation.SuppressLint
import android.app.AlertDialog
import android.app.DatePickerDialog
import android.app.TimePickerDialog
import android.icu.text.DateFormat
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.text.format.DateFormat.is24HourFormat
import android.widget.*
import java.util.Calendar
```

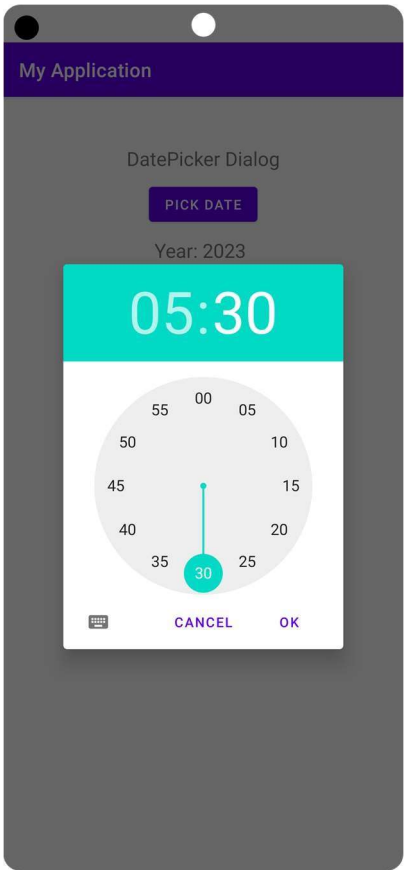
```
class MainActivity : AppCompatActivity(), DatePickerDialog.OnDateSetListener,
TimePickerDialog.OnTimeSetListener {

    lateinit var btnPickDate: Button
    lateinit var tvDate: TextView
    lateinit var btnPickTime: Button
    lateinit var tvTime: TextView
    lateinit var btnClickMe: Button
    lateinit var tvAlertDialog: TextView
    var day: Int = 0
    var month: Int = 0
    var year: Int = 0
    var hour: Int = 0
    var minute: Int = 0
    var myDay: Int = 0
    var myMonth: Int = 0
    var myYear: Int = 0
    var myHour: Int = 0
    var myMinute: Int = 0


    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        btnPickDate = findViewById(R.id.btnPickDate)
        tvDate = findViewById(R.id.tvDate)
        btnPickTime = findViewById(R.id.btnPickTime)
        tvTime = findViewById(R.id.tvTime)
        btnClickMe = findViewById(R.id.btnClickMe)
        tvAlertDialog = findViewById(R.id.tvAlertDialog)

        btnPickDate.setOnClickListener {
            val calender = Calendar.getInstance()
            day = calender.get(Calendar.DAY_OF_MONTH)
            month = calender.get(Calendar.MONTH)
            year = calender.get(Calendar.YEAR)
            val datePickerDialog = DatePickerDialog(this@MainActivity, this@MainActivity,
year, month, day)
            datePickerDialog.show()
        }

        btnPickTime.setOnClickListener {
            val calender = Calendar.getInstance()
            hour = calender.get(Calendar.HOUR)
            minute = calender.get(Calendar.MINUTE)
            val timePickerDialog = TimePickerDialog(this@MainActivity, this@MainActivity,
hour, minute, true)
            timePickerDialog.show()
        }

        btnClickMe.setOnClickListener {
            val builder = AlertDialog.Builder(this)
            builder.setTitle("Dialog Box")
            builder.setMessage("Deleting File may harm your system")
```

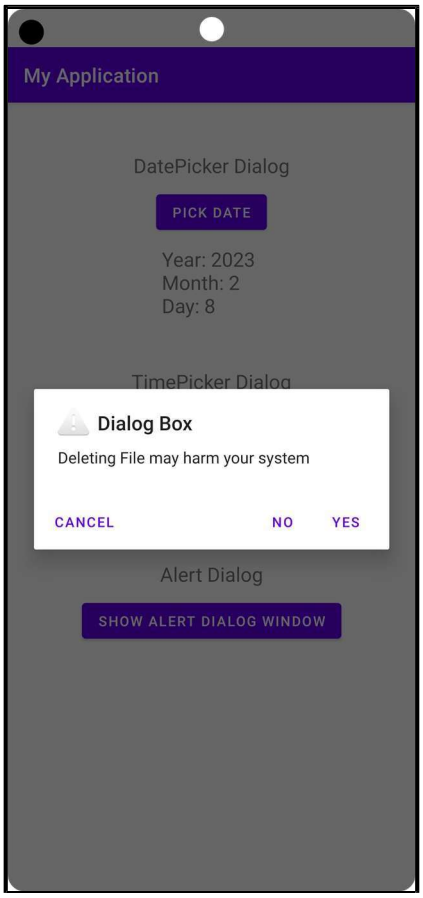
```
            builder.setIcon(android.R.drawable.ic_dialog_alert)

            builder.setPositiveButton("Yes"){dialogInterface, which ->
                tvAlertDialog.text = "Clicked Yes"
            }

            builder.setNegativeButton("No"){dialogInterface, which ->
                tvAlertDialog.text = "Clicked No"
            }

            builder.setNeutralButton("Cancel"){dialogInterface , which ->
                tvAlertDialog.text = "Clicked Cancel\nOperation Cancelled"
            }

            val alertDialog: AlertDialog = builder.create()
            alertDialog.setCancelable(false)
            alertDialog.show()
        }
    }

    override fun onDateSet(p0: DatePicker?, p1: Int, p2: Int, p3: Int) {
        myYear = p1
        myMonth = p2+1
        myDay = p3
        tvDate.text = "Year: " + myYear + "\n" + "Month: " + myMonth + "\n" + "Day: "
+ myDay
    }

    override fun onTimeSet(p0: TimePicker?, p1: Int, p2: Int) {
        myHour = p1
        myMinute = p2
        tvTime.text = "Hour: " + myHour + "\n" + "Minute: " + myMinute
    }
}
```

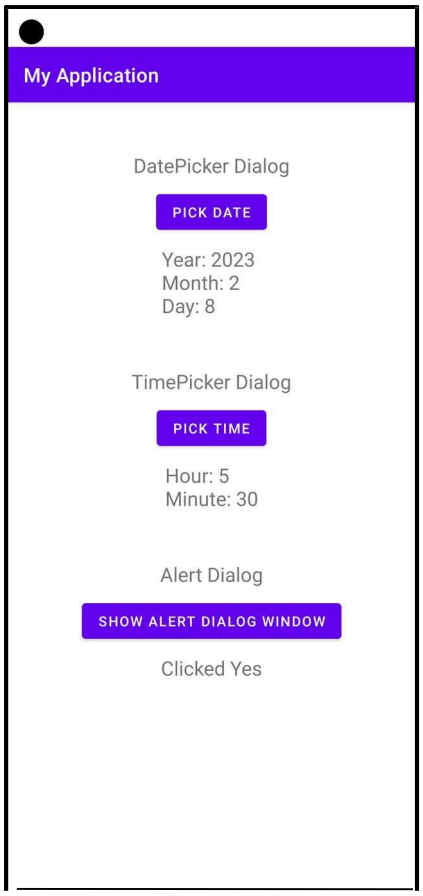
Output


My Application
DatePicker Dialog
PICK DATE
TimePicker Dialog
PICK TIME
Alert Dialog
SHOW ALERT DIALOG WINDOW


My Application
DatePicker Dialog
PICK DATE
2023
Wed, Feb 8
February 2023
S M T W T F S
CANCEL
OK


My Application
DatePicker Dialog
PICK DATE
Year: 2023
05:30
CANCEL
OK


My Application
DatePicker Dialog
PICK DATE
Year: 2023
Month: 2
Day: 8
Dialog Box
Deleting File may harm your system
CANCEL
NO
YES
Alert Dialog
SHOW ALERT DIALOG WINDOW


My Application
DatePicker Dialog
PICK DATE
Year: 2023
Month: 2
Day: 8
TimePicker Dialog
PICK TIME
Hour: 5
Minute: 30
Alert Dialog
SHOW ALERT DIALOG WINDOW
Clicked Yes